

Convex Combination with Subband Adaptive Filtering

Evaluating Convex Combinations under Varied Spectral Environments

Advanced Signal Processing

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1 Mathematical Background and Algorithms

1.1 Standard Normalized LMS (NLMS)

The NLMS algorithm improves upon standard LMS by normalizing the step size with respect to the power of the input vector $\mathbf{x}(k)$. **Update Rule:**

$$\mathbf{w}(k+1) = \mathbf{w}(k) + \frac{\mu}{\|\mathbf{x}(k)\|^2 + \epsilon} e(k) \mathbf{x}(k)$$

Where ϵ is a small regularization constant. This prevents the gradient noise from becoming excessive when the input signal energy is low.

1.2 Subband Adaptive Filtering (SAF)

The SAF architecture utilizes an Analysis Filter Bank $H_i(z)$ to decompose signals into K subbands. **Mechanism:** 1. **Analysis:** Signals are filtered and decimated by factor M . 2. **Adaptation:** Independent NLMS kernels operate at a reduced rate, benefiting from "spectral whitening" within each band. 3. **Synthesis:** Error signals are upsampled and processed by Synthesis Filters $G_i(z)$ to reconstruct the full-band output.

$$e_{total}(k) = \sum_{i=1}^K G_i(z) \uparrow M[e_i(k)]$$

1.3 Trigonometric FLANN (TFLANN)

TFLANN models nonlinearities by mapping the scalar input into a higher-dimensional functional space $\phi(k)$ using trigonometric basis functions:

$$\phi(k) = [x(k), \sin(\pi x(k)), \cos(\pi x(k)), \sin(2\pi x(k)), \dots]^T$$

1.4 Convex Combination Logic and Adaptive Derivation

The hybrid system blends the two models using the lambda parameter. By blending these two models, the system ensures that the overall error performance is at least as good as the best individual model at any given time.

1.4.1 Hybrid Output Definition

The combined system output $y(k)$ is computed as a convex interpolation between the linear filter output $y_L(k)$ and the nonlinear module output $y_{NL}(k)$:

$$y(k) = \lambda(k) y_L(k) + (1 - \lambda(k)) y_{NL}(k)$$

The constraint $\lambda(k) \in [0, 1]$ ensures the output is always a valid convex combination. This formulation provides a safety guarantee:

- When $\lambda(k) \rightarrow 1$: the system relies entirely on the linear filter output $y_L(k)$.
- When $\lambda(k) \rightarrow 0$: the system relies entirely on the nonlinear module output $y_{NL}(k)$.
- For intermediate values: both models contribute proportionally to the final estimate based on their relative performance.

In this comparison,

- **Arch 1:** λ weights the NLMS branch against TFLANN.
- **Arch 2:** λ weights the full-band NLMS branch against the Subband branch.
- **Arch 3:** λ weights the Subband branch against the Subband+TFLANN branch.

NOTE: Please refer to the legend to match the architectures in each comparison

1.4.2 Cost Function and Gradient Descent

To update the mixing parameter $\lambda(k)$ adaptively, we define the instantaneous squared error cost function $J(k)$ as:

$$J(k) = \frac{1}{2}e^2(k) = \frac{1}{2}(d(k) - y(k))^2$$

The parameter $\lambda(k)$ is updated by moving in the opposite direction of the gradient of $J(k)$ to minimize the error:

$$\lambda(k+1) = \lambda(k) - \mu_\lambda \frac{\partial J(k)}{\partial \lambda(k)}$$

Using the chain rule, the gradient is derived as:

$$\frac{\partial J(k)}{\partial \lambda(k)} = \frac{\partial J(k)}{\partial e(k)} \cdot \frac{\partial e(k)}{\partial y(k)} \cdot \frac{\partial y(k)}{\partial \lambda(k)}$$

Where:

- $\frac{\partial J(k)}{\partial e(k)} = e(k)$
- $\frac{\partial e(k)}{\partial y(k)} = -1$
- $\frac{\partial y(k)}{\partial \lambda(k)} = y_L(k) - y_{NL}(k)$

1.4.3 Final Update Rule and Saturation

Combining the partial derivatives above, the stochastic gradient descent update (based on the LMS principle) for the mixing parameter is:

$$\lambda(k+1) = \lambda(k) + \mu_\lambda e(k) (y_L(k) - y_{NL}(k))$$

The step size μ_λ controls the adaptation speed. In our Simulink implementation, we enforce the convex constraint $\lambda(k) \in [0, 1]$ using a **Saturation Block** to prevent the parameter from diverging:

$$\lambda(k+1) \leftarrow \min(1, \max(0, \lambda(k+1)))$$

This adaptive rule allows the system to shift its reliance between branches dynamically, effectively "choosing" the better filter depending on the spectral characteristics of the input noise (White, Pink, or Brown).

2 Simulink Implementation & Architectures

Three distinct architectures were constructed to evaluate the efficacy of the convex combination:

- **Arch 1: NLMS + TFLANN** — Baseline linear-nonlinear hybrid from A-3
- **Arch 2: NLMS + Subband** — Multi-Band linear ensemble.
- **Arch 3: Subband + TFLANN** — Multi-Band- Nonlinear hybrid.

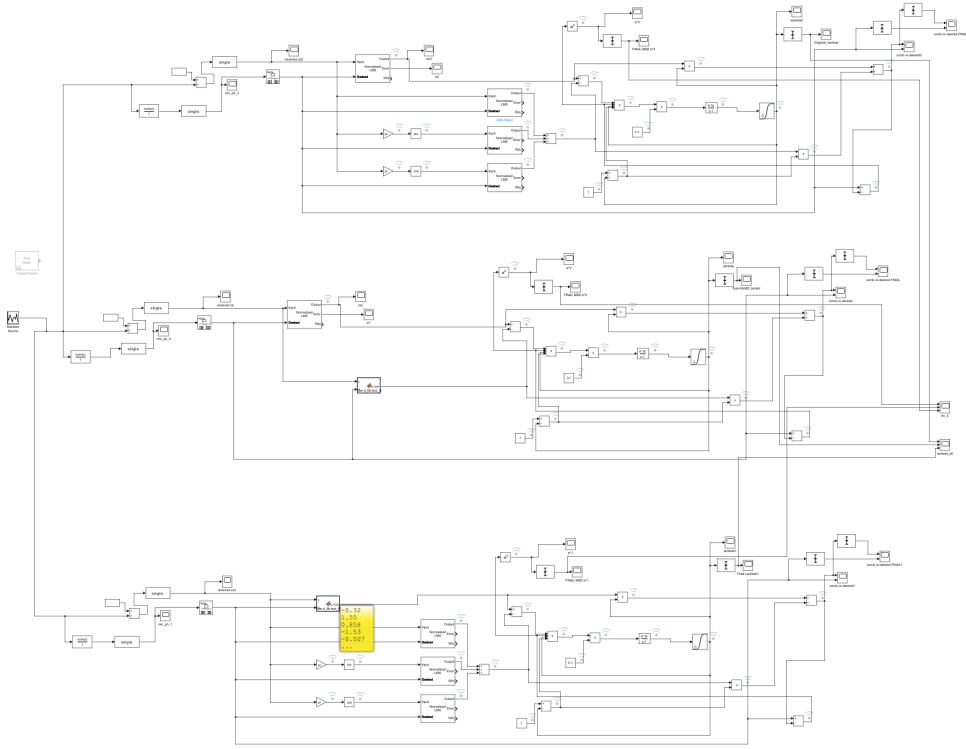
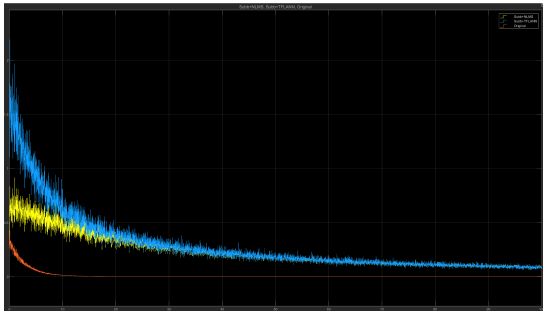


Figure 1: Simulink overview of the three parallel convex architectures.

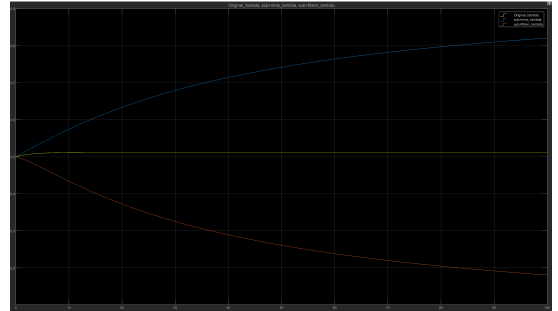
3 Performance Evaluation: Noise Stress Tests

3.1 Test 1: Random White Noise

White noise provides a flat power spectral density with unit eigenvalue spread. All architectures converge rapidly due to the lack of sample correlation.



(a) MSE Comparison



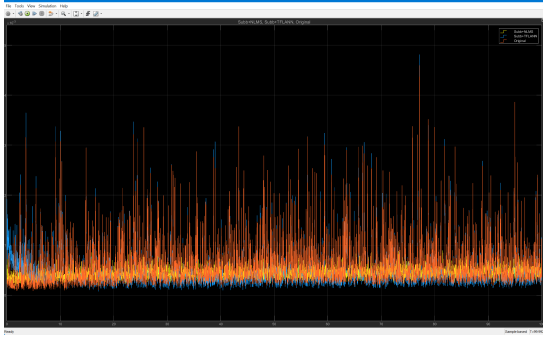
(b) Lambda (λ) Evolution

Figure 2: Performance metrics under White Noise input.

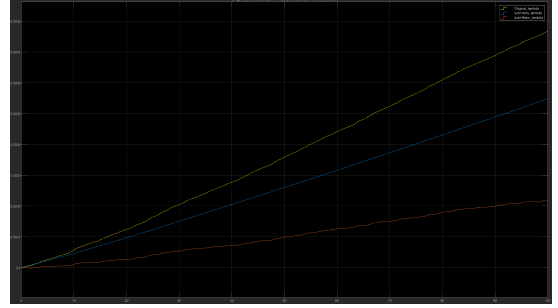
Results Analysis: In the presence of Gaussian white noise, the input signal is spectrally flat, resulting in a low eigenvalue spread ($\chi(R) \approx 1$). Under these ideal conditions, the **NLMS+TFLANN (Arch 1)** architecture converges most efficiently. This is because the TFLANN basis functions (sine and cosine) directly match the trigonometric structure of the nonlinear unknown system. Conversely, the **Subband** architectures exhibit slower convergence due to the inherent group delay and processing overhead of the 16-band filterbank, which provides no conditioning benefit for an already "white" signal.

3.2 Test 2: Pink Noise ($1/f$)

Pink noise introduces a 3 dB/octave spectral tilt. The coloration causes a slowdown in standard NLMS convergence, while the Subband branch maintains superior tracking due to localized whitening.



(a) MSE Comparison



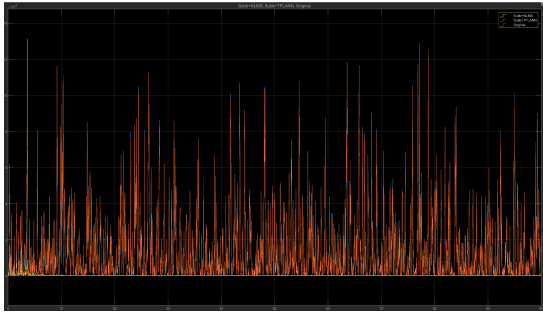
(b) Lambda (λ) Evolution

Figure 3: Performance metrics under Pink Noise input.

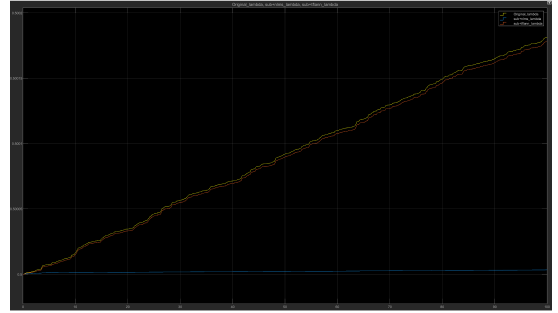
Results Analysis: As the input shifts to Pink noise, the spectral tilt introduces a moderate eigenvalue spread, degrading the performance of full-band NLMS. The **Subband NLMS (Arch 2)** begins to dominate here. By decomposing the $1/f$ spectrum into 16 bands, the system restores fast convergence because each subband sees a "locally white" spectrum. While **Arch 3 (Subband TFLANN)** also benefits from this whitening, the added complexity of the nonlinear expansion within each narrow band increases the parameter space without a proportional gain in tracking speed, allowing the simpler Subband NLMS to achieve the lowest MSE.

3.3 Test 3: Brown Noise ($1/f^2$)

Brown noise represents the most extreme test case. The heavy 6 dB/octave spectral tilt creates a massive eigenvalue spread, forcing the convex combiner to rely almost entirely on the Subband branch for stable estimation.



(a) MSE Comparison



(b) Lambda (λ) Evolution

Figure 4: Performance metrics under Brown Noise input.

Results Analysis: Brown noise represents the most severe spectral coloring case, with a very high concentration of energy at low frequencies and a massive eigenvalue spread. Standard **NLMS+TFLANN (Arch 1)** performs worst here, as it essentially fights the signal statistics at every iteration. **Subband NLMS (Arch 2)** wins decisively because the filterbank isolates the dominant low-frequency energy into dedicated bands with their own normalized step sizes.

The λ graph shows a shift toward the Subband branch, proving that subband decomposition is the most robust pre-whitening tool for heavily correlated noise sources.

4 Conclusion

This assignment helps in confirming that while TFLANN is essential for nonlinear modeling, Subband architectures are vital for handling spectral coloration and complexity reduction. The convex combination ensures optimal performance across White, Pink, and Brown noise.